

Rocksteady research report — nephrotoxicity

Responding to: toxicity/kidney/BEEB0P_RESEARCH_REQUEST_2026-10-02.md (review snapshot 6c5797a2590e) **Branch:** claude/amazing-galileo-rwiv95 · **Inventory commit:** b51003c02ef66b5f226dff02300e3970367a855b **Dataset version:** release kidney-224e9a2f7916 — 246 measurements · 65 roster entries · 42 clinical register rows **Research date:** 2026-10-02 · **Authorization:** Oscar, 2026-10-02, research/review round **Status:** RESEARCH ROUND. No validated data, label, grade, eligibility class, adjudication or model was changed. Acquired material is held in research_staging/, outside data/.

Work actually performed this round: **7 primary sources retrieved and read in full** (4 trial reports, German's routed human 3D-PTEC paper, Moisan 2017 + its supplement, Crooke 2018), **20/20 resources logged, 36 trial registry records surfaced, 6 of 10 missing sequences recovered**. Work proposed but not performed is marked as such throughout.

REVISION B — 2026-10-03 (later)

ENVISION is retrievable. I was wrong to leave it on the inaccessible list, and I was wrong twice: once by not finding the route, and once by dismissing it when a cross-check found it for me.

Revision A's §P5 note records that a cross-checking agent claimed ENVISION was free from an institutional repository, that I tested the route and could not substantiate it, and that ENVISION therefore stayed on the access list. **That dismissal was unsound.** What I actually tested was Semantic Scholar's advertised PDF (which resolves to nejm.org and is Cloudflare-blocked) and CORE (403), while OpenAlex — the service that enumerates repository copies — was rate-limited, so I never enumerated the open locations at all. I generalised "the routes I could test failed" into "the claim is unsubstantiated". When the completed cross-check supplied an exact URL, it worked on the first attempt:

kclpure.kcl.ac.uk/ws/files/148225585/nejmoa1913147.pdf — **HTTP 200, 621,752 B, 14 pp, 9,651 words, "Publisher's PDF, also known as Version of record"**, SHA-256 6c54ccd57669ede6.... King's Research Portal. Staged.

Retrievable trials are 4 of 10, not 3 — and ENVISION is the renal-richest of the ten.

What ENVISION actually reports (NCT03338816, N=94; safety population 46 placebo / 48 givosiran)

Note the denominator trap: the paper also prints N=43 / N=46, which is the **acute intermittent porphyria efficacy subgroup**, not the safety population. The two must not be conflated.

Renal endpoint	Placebo (n=46)	Givosiran (n=48)
Any renal adverse event	3 (7%)	7 (15%)
Increased serum creatinine or decreased eGFR	2 (4%)	7 (15%)
Chronic kidney disease	0	5 (10%)
Decreased eGFR	0	3 (6%)
Worsening CKD as a serious adverse event	0	2

Reporting threshold: renal AEs were defined as all events under **MedDRA terms for chronic kidney disease**; the AE table lists events ≥5 percentage points higher in one group. Baseline: **34% of patients had eGFR < 60 mL/min/1.73 m²**, and 25% had renal impairment in their medical history. The abstract states plainly that the increased efficacy *"was accompanied by a higher frequency of hepatic and renal adverse events."*

This confirms MSR009 (readout_value = 15 pct_incidence) against the version of record and adds the denominators the row lacks. It is a **non-renal indication**, so it escapes the renal-indication confound of §P3 — but it carries a different one that must be recorded rather than ignored: acute hepatic porphyria is itself associated with

chronic kidney disease, and the two patients with worsening CKD had **renal-biopsy results "consistent with their underlying coexisting illnesses (hypertension and porphyria-associated nephropathy)"**. The signal is real and drug-associated at the adverse-event level; the paper does not attribute it to direct tubular toxicity. Both facts belong in the row.

Two further corrections to §5 and §6

- **The Europe PMC** supplementaryFiles **endpoint works** and I had not used it for trials. PMC8487715 returns a 1,706,506 B archive containing cir-144-1133-s001.pdf (663,816 B) — the teprasiran supplementary appendix, which holds the RIFLE/KDIGO and MAKE90 tables cited as "Table IV/V in the Data Supplement". It works only for open-access records, so it does **not** reach the vupanorsen supplement.
- **Vupanorsen is not paywalled content.** Crossref reports 10.1161/CIRCULATIONAHA.122.059266 licensed **CC BY-NC-ND 4.0** from 2022-05-03. Its 403 is a Cloudflare bot wall on openly licensed material. Access item 4 in §5 should read **openly licensed, technically blocked** — a browser, not an entitlement.

Revised verified-trial trajectory: 2 → 6 is now plausible (PROMOVI, teprasiran, PHYOX3, ENVISION promotable on evidence in hand, subject to German), not the 2 → 5 I projected.

Method note I am keeping. Verifying each cross-check before acting was right — the same agent's earlier, vaguer version of this claim was correctly held back, and a different agent's "green OA" leads for mongersen (Tor Vergata, login wall) and donidalorsen (Jagiellonian, metadata-only record) are genuinely false. The error was not scepticism; it was reporting "unsubstantiated" when the honest statement was "I could not test the route that matters, because OpenAlex was rate-limited."

REVISION A — 2026-10-03

Two findings in §P4 below were wrong and are corrected here. The original text is left intact beneath, annotated, rather than edited away.

A1. Sequence recovery: 6 of 10 recovered, not 0 — and my reasoning was the exact error I had just warned against

§P4 states that the four Moisan AON sequences are "verified unpublished" because I read the paper in full and found zero nucleotide strings. **That conclusion was wrong.** The strings are absent from the article body because they are in **Table S2 of the supplement**, which the article itself points to: "*Sequences and backbone modifications are provided in Table S2.*" I read the article and did not fetch the supplement — the precise article-versus-supplement distinction that §P5 of this very report argues for. The failure was mine, not the source's.

The supplement is free: europepmc.org/.../PMC5363415/supplementaryFiles returns HTTP 200, 8.47 MB. Table S2's body is a **raster image with no text layer** — page 18 of mmc1.pdf yields 33 words, all caption and legend — so it defeats text extraction. Rendered at 400–600 dpi and read visually.

Separately, ISIS 113715 and ISIS 104838 are in **Crooke 2018 Table 1**, machine-readable, with <underline> markup giving per-position 2'-MOE wings. **That source was already cited in our own oligos.csv as OLG031's design_source.** I had searched for the compound names instead of reading the source our dataset already pointed at.

Oligo	Recovered sequence	Source
OLG025 ISIS113715	GCTCcttccactgatCCTGC (20 nt, PTP1B)	Crooke 2018 Table 1
OLG026 ISIS104838	GCTGAttagagagagGTCCC (20 nt, TNFA)	Crooke 2018 Table 1
OLG041 AON-A	mCGTcagtatgcgAATc (16 nt, scramble control)	Moisan Table S2
OLG042 AON-C	GmCtgtgtgagcttGG (15 nt, PCSK9)	Moisan Table S2
OLG043 AON-D	AATgctacaaaamCmCmCA (16 nt, PCSK9)	Moisan Table S2

Oligo	Recovered sequence	Source
OLG044 MYD88-E	TAAggcaatcaagGTA (16 nt, MyD88)	Moisan Table S2

Validation, because a wrong sequence is worse than a missing one. The Crooke parse was checked against three sequences we already hold by an independent route: mipomersen and volanesorsen match base-for-base; pelacarsen matches after U→T normalisation, which incidentally exposes a **notation inconsistency in our own row** (UGCUCGTTGGTGCTUGUUC mixes U and T for a DNA gapmer). The Moisan read is validated by row B, which reproduces OLG002 SPC5001 exactly once methyl marks are stripped — independently confirming AON-B = SPC5001.

Two bonus per-position chemistry upgrades fall out: Crooke supplies 5-10-5 wing encoding for OLG006 mipomersen and OLG005 volanesorsen, which our all-uppercase rows lack.

Revised coverage: 61/63 molecules (96.8%), not 55/63. The two genuinely remaining are OLG014 and OLG015 (Sandelius 2020, SAGE returns 403 on all three routes; isOpenAccess=N). OLG030 and OLG031 remain class aggregates and still should not sit in the denominator.

All six are staged as **proposals pending German** in research_staging/sequences_recovered_2026-10-02.csv with per-row locus, retrieval route, extraction method and transcription confidence. **Nothing was written to** data/oligos.csv. The four Moisan transcriptions are marked medium confidence and want a second reader, since they came from an image; OLG041's trailing residue is recorded as the lowercase c that is printed and is flagged.

A2. Bridge unit-match is 0/8, not 3/8 — the finding is stronger than I published

The unit shared 3/8 figure carried from my October 1 reply **does not reproduce**. Recomputed over the argmax rows that human_animal_bridge.csv actually names: **2/8 counting NA as a match, and 0/8 excluding it**. The two apparent matches are OLG012 and OLG013, both of which are NA on both sides — the absence of a unit, not agreement on one. My 3/8 came from widening to all rows for the oligo and counting NA == NA.

Worse, **the one genuine endpoint match fails on units too**. OLG008 compares MSR018 human urinary A1M at **36.4 mg/L** against MSR020 monkey urinary A1M at **21 fold_change** — same analyte, non-comparable scale. My reply also named the wrong human row for it (MSR023, an in-vitro uptake assay); the endpoint match actually comes from the clinical row MSR018.

So the corrected bridge position: **1/8 share an endpoint, 5/8 share a route, 0/8 share a comparable unit, and the single endpoint match is not quantitatively comparable**. Zero of eight "paired" comparisons support a quantitative translational claim. That strengthens the recommendation to retire comparison_type rather than weakening it.

One consequence needs flagging to German and Oscar directly: the only animal_over_predicts row in the entire bridge is OLG042, and OLG042 fails endpoint, unit **and** route matching simultaneously — it is the worst-matched of the eight. PRESENTATION.md still headlines "animal tests over-predict human risk". **The most externally visible claim in the project rests on its least defensible comparison**. This is now the highest-priority documentation fix, ahead of the items in my October 1 reply.

1. Proposal dispositions

#	Proposal	Disposition
1	Prioritize the 67 human laboratory rows; investigate 10.2131/jts.51.75	ACCEPT — substantially executed
2	Verify the remaining 10 clinical studies, starting with teprasiran / PHYOX3 / PROMOV	ACCEPT — 4 of 10 executed (3 here + ENVISION in Revision B)
3	Reconsider MSR066	ACCEPT — and the defect is four times larger than stated

#	Proposal	Disposition
4	Per-dimension bridge audit; recover the 10 missing sequences	MODIFY — audit already complete; recovery target is not 10
5	Correct access classifications	ACCEPT — executed, and it immediately paid off

P1 — Human laboratory rows and the routed 3D-PTEC source — ACCEPT, substantially executed

German's routed source is **Morimura et al., Predicting nucleic acid drug-induced nephrotoxicity using a 3D human renal proximal tubule spheroid model, J Toxicol Sci 51(1):75–87, 2026** (10.2131/jts.51.75, PMID 41500580). Europe PMC reports `isOpenAccess=N, inEPMC=N`. **It is nevertheless free:** J-STAGE serves the full PDF at `jstage.jst.go.jp/article/jts/51/1/51_75/_pdf/-char/en`. Retrieved, 13 pp, 6,530 words, SHA-256 2882260fdfa7dcc9.... This single case vindicates P5 before P5 is even argued: a PMC "not open access" flag said nothing about the publisher's own free route.

Overlap check, as requested before staging — the answer is complete overlap, and that is the good outcome. The paper tests exactly three compounds, all already on our roster:

Test compound	Our id	Chemistry (per Table 1)	MW (g/mol)	Vendor / catalog	Cmax (Table 2)
SPC5001	OLG002	PS, LNA, 5mC	4689.85 (5726.02 Alexa-647 labelled)	Ajinomoto Bio-Pharma Services, custom synthesis	unknown
Viltolarsen	OLG013	PMO	7386.42	MedChemExpress HY-132586A	0.043 µmol/L
Givosiran	OLG003	PS, 2'-OMe, 2'-F	16300.6	MedChemExpress HY-132610	0.021 µmol/L

Why this matters more than a new-compound find:

- It is a human-laboratory ↔ human-clinical bridge for three compounds.** All three already carry human *clinical* renal outcomes in our dataset. That is a human-to-human comparison, which is strictly better evidence than the animal bridge whose eight "pairs" collapsed to one matched comparison in the October 1 review.
- It independently corroborates our strongest translational finding.** The paper's own Table 2 records SPC5001 as "withdrawn due to nephrotoxicity... multiple foci of tubular necrosis and oligonucleotide deposition... in the renal biopsy" while its nonclinical column reads "no clear nephrotoxicity was observed". That is the animal-under-predicts case, stated by an independent group, and the 3D-RPTEC model detects SPC5001 ("significant ATP depletion... only after prolonged exposure to SPC5001").
- It is the first concrete lead on the 0/65 purity gap.** Two of three test articles are catalogue items with vendor part numbers. A vendor certificate of analysis is a *tested-material* purity document, which is precisely the artefact the Challenge asks for and that no trial paper or label publishes. This does not close the gap — the vendor's lot is not the trial's lot, and that distinction must be preserved — but it converts "unobtainable" into "obtainable for a defined subset", which is a different problem.
- Readouts are ATP, LDH, KIM-1, NGAL and high-content analysis, directly commensurate with our existing `intracellular_ATP` and `KIM-1` human in-vitro readouts.
- It repairs one of the two weakest bridge pairs identified on October 1.** OLG013 is viltolarsen, and its human side is currently `kidney_toxicity_monitored` — a monitoring assertion, not a measurement, which is exactly why I flagged that pair as uninterpretable. Morimura supplies real human in-vitro readouts for viltolarsen, and its Table 2 records clinical **increased β2-microglobulin and NAG** plus nonclinical increased BUN/creatinine. That converts OLG013 from an assertion-versus-lesion comparison into a measurement-versus-measurement one.

Current state of the 67 human laboratory rows (computed at b51003c, not asserted): 55/67 carry a numeric `readout_value`, 61/67 carry a dose or concentration. The gap P1 targets is therefore not primarily numerical-readout recovery — it is **per-position chemistry, donor/model context and tested-material method**,

which is where the Morimura table contributes and where our patent-derived rows remain thin.

Not performed: extraction of the paper's per-compound dose–response values into staged rows. The figures are the quantitative locus (Figs 3–6) and figure-level extraction from a PDF needs the anchor-check discipline we use for patent tables; I did not attempt it in the time available and would rather do it correctly than quickly.

P2 — Clinical study verification — ACCEPT, 3 of 10 executed (4 of 10 after Revision B)

All three leads named in my October 1 reply were retrieved via the Europe PMC REST `fullTextXML` endpoint and read in full. Checksums and HTTP evidence in `research_staging/logs/acquisition_manifest.json`; extracted facts in `research_staging/trial_verification_2026-10-02.csv`.

PROMOVI / eteplirsen (MSR040) — CONFIRMED AND ENRICHED. NCT02255552, N=79, open-label, 96 weeks. Source: "Overall, 8 eteplirsen-treated patients (10.1%) experienced renal TEAEs; each as proteinuria, which resolved by end of study in all but one individual." Our row already said proteinuria 10.1 pct_incidence, grade 1 — **correct, and now source-verified rather than search-derived.** The paper adds what the row lacked: the denominator (8/79), and that **3/79 (3.8%) of the proteinuria was investigator-assessed as drug-related** (Table 5). This is our single best clinical row — human, quantified, denominator-bearing, and in a **non-renal indication**, so it is not subject to the confound in P3. It would be the **third verified trial**.

Teprasiran (MSR077) — quantitative value recovered, confound identified. NCT02610283, 360 randomised, 341 dosed, mITT-bsCr 165 placebo / 157 teprasiran. Monitoring: daily to day 7, then ~day 30/90/365, serum creatinine and cystatin C, independent DMC. Thresholds: AKIN on sCr through day 5; MAKE90 = death, RRT, or ≥25% eGFR reduction at day 90. Results: AKI 58/157 (36.9%) vs 82/165 (49.7%), ARR –12.8%, OR 0.58, p=0.02 — **this is the efficacy endpoint.** The separable safety comparisons are AKI-as-SAE 5.5% vs 6.8%, and MAKE90 20.9% teprasiran vs 19.3% placebo, OR 1.1 (95% CI 0.6–1.9), p=0.71. Investigator-attributed related AEs were hypotension, ALT and AST — none renal. Our row's AKI_incidence = reduced has no number; the number now exists, but see P3: the primary endpoint is therapeutic, so the defensible locus is the safety comparison, not the headline.

Extension-counting, which P2 explicitly asked for. The teprasiran paper names **four** NCTs: NCT02610283 (this phase 2), NCT03510897 (the larger phase 3 built on it), and NCT00802347 / NCT02610296 (delayed graft function in kidney transplant). Registry search separately surfaced NCT00554359, a phase 1 of **ISNP** — an earlier name for the same molecule. Five trial identifiers, one molecule. Our register currently holds one teprasiran study key, which is correct, but the convention needs stating explicitly before any of the other four is added, or the trial total will inflate on a single drug.

PHYOX3 / nedosiran (MSR046) — PARTIALLY CONFIRMED, and a provenance split is required.

NCT04042402 (rollover from NCT03847909), N=13, interim analysis at 2.5 years. The paper's primary objective — annual rate of eGFR decline — was *too preliminary to analyse*; only descriptive eGFR is reported, as "mean eGFR remained stable (62–84.2 mL/min per 1.73 m²) to month 30". **Our row's value** eGFR = 2.5 pct_change **does not appear in this paper** and appears to derive from the Rivfloza label, which the source_ref lists alongside it. One row currently cites two documents for one number without saying which supplied it. Also material and absent from our row: **3/13 (23.1%) had serious adverse events including acute kidney injury, kidney failure and pyelonephritis** — all investigator-assessed as unrelated to nedosiran, because kidney stones and nephrocalcinosis are PH1 disease manifestations.

Not performed: the remaining 7 of 10 (6 after Revision B — *ENVISION* was retrieved). Four have no PMC deposit at all (donidalorsen, ENVISION givosiran, mongersen, OCEANa-DOSE olpasiran — all NEJM); two are in Europe PMC but not open access and are bot-walled (NEURO-TTR inotersen PMC12611561, van Poelgeest SPC5001 PMC4693495); one (vupanorsen) has its article free but its renal numbers in a 403 supplement. See §5.

P3 — MSR066 — ACCEPT, and the defect is four times larger than Beebop found

Beebop is factually right. MSR066 records cemdisiran in IgA nephropathy with effect_vs_control = "placebo –6.3; cemdisiran –2.9 (favourable)", readout_value = –2.9 mL/min/1.73 m² change from baseline at week 32, and negative_eligibility = confirmed_negative. Both arms are losing filtration; the drug arm loses less. That is a **therapeutic effect on kidney-disease progression**, and it cannot establish absence of nephrotoxicity, because a therapeutic benefit and a toxic injury act on the same measurement in opposite directions and are not separable from a single between-arm difference. Our own CLINICAL_VALIDATION.md:51

contains the tell in writing: "A genuine measured negative, **in fact favourable vs placebo.**"

The source settles it in its own words. I retrieved the cemdisiran report (PMC11020434, 121,950 B) and read its End Points section. The primary end point is percentage change from baseline in **urine protein-to-creatinine ratio**; the secondary end points are UPCR variants and clinical remission. Of the week-32 eGFR change that MSR066 records, the paper says:

"Exploratory end points included change from baseline in eGFR at week 32 and eGFR slope over the course of the study."

and, of its statistics:

"No formal statistical hypothesis testing was performed, and no multiplicity adjustments were made."

So the number behind our only confirmed_negative is a **prespecified exploratory endpoint, inside the efficacy family, with no hypothesis testing and no multiplicity adjustment**. It is not a safety measurement by the source's own framing, and it carries no inferential weight even as efficacy. That is three independent reasons it cannot serve as a toxicity-negative control, and it makes the reclassification below a correction rather than a judgment call.

The independent finding: this is not one mislabelled row, it is an unruled class. Four of our 42 clinical rows are drugs whose *indication is itself a kidney disease*. All four are grade 0. All four are classified differently:

Row	Drug	Indication	negative_eligibility
MSR045	lumasiran	primary hyperoxaluria type 1	efficacy_derived_negative
MSR046	nedosiran	primary hyperoxaluria	asserted_negative_regulatory
MSR066	cemdisiran	IgA nephropathy / PNH	confirmed_negative
MSR077	teprasiran	acute kidney injury prevention	not_eligible_negative

Same structure — renal-indication drug, renal functional endpoint, renal-disease population — four different answers. MSR045 already carries the right one, so the dataset contains its own precedent.

Source-grounded disposition for German, no label changed: introduce a *derived* rule — if the oligo's indication is a renal disease **and** the readout is a renal functional endpoint, then negative_eligibility = efficacy_derived_negative and nephrotox_grade_modeling is left blank. Derived, not hand-entered, consistent with how subject_class and renal_endpoints_measured are already produced. Applied, it reclassifies MSR046, MSR066 and MSR077 and leaves MSR045 as-is.

Consequence German must weigh, stated plainly: MSR066 is the **only** confirmed_negative among 42 clinical rows. Reclassifying it leaves the kidney dataset with **zero confirmed clinical negatives**. I think that is the correct and more honest state — and it converges with German's independent finding of zero clean sequence-linked clinical negatives in thrombocytopenia, which suggests a property of the evidence base rather than a curation failure in either endpoint. It also means no clinical row is currently usable as a modeling negative, which is a material fact for the narrative and for any future classifier.

Note that MSR040 (etelplirsen, DMD) is untouched by this rule and remains a measured human clinical **positive** — the non-renal indications are where clean nephrotoxicity signal lives.

P4 — Bridge audit and sequence recovery — MODIFY

The per-dimension bridge audit was completed and published on October 1, before this request. Its verified figures: of 8 paired_same_source comparisons, **1/8 share a biological endpoint** (OLG008; the match is the clinical row MSR018 vs MSR020, **not** the in-vitro A1M/RAP uptake row as my October 1 reply said), **~3/8 share a unit** — **0/8 share a comparable unit** — **see Revision A2**, 5/8 share a delivery route, 2/8 compare a monitoring assertion (kidney_toxicity_monitored) against an observed animal lesion, and the pairing test is **vacuous where grades tie at 0** because the argmax set becomes the whole row set. Recommendation stands: retire comparison_type in favour of per-dimension match flags and label the bridge hypothesis-generating. Beebop's

framing — "eight same-source links and one shared analyte do not establish eight comparable translational pairs" — matches that result exactly.

Sequence recovery: the target is not ten. Recovered this round: zero. Honest breakdown:

SUPERSEDED BY REVISION A1 (2026-10-03). Six of ten were recovered. The "verified unpublished"

finding below is wrong: the Moisan sequences are in the supplement's Table S2, which I did not

fetch. The table below is retained as the original record.

Count	Entries	Status
2	OLG030 GalNAc_siRNA_class_Janas, OLG031 ASO_2MOE_class_human_pooled	Not molecules. Class-level aggregates pooling observations across a chemistry class. They can never carry a sequence and should not sit in a sequence-coverage denominator.
4	OLG041 AON-A, OLG042 AON-C, OLG043 AON-D, OLG044 MYD88-E	~~Verified unpublished.~~ WRONG — see Revision A1. The article body contains zero nucleotide strings because the sequences are in supplementary Table S2, which is free and which I failed to fetch. All four recovered.
2	OLG025 ISIS113715, OLG026 ISIS104838	~~Not recoverable from open literature.~~ WRONG — see Revision A1. Both are in Crooke 2018 Table 1, a source already cited in our own oligos.csv. Recovered. No patent search needed.
2	OLG014, OLG015 cEt tool/control ASOs	Source is Sandelius 2020 (PMID 33084520), not retrieved this round. Tool compounds; disclosure uncertain.

Denominator correction: sequence coverage should be reported over 63 molecules, not 65, because two roster entries are class aggregates. **Revised figure after Revision A1: 61/63 (96.8%).** The 87.3% stated here is superseded.

Dedup check, unprompted and clean: Moisan 2017 states "a PCSK9-targeting LNA-AON drug (SPC5001, herein AON-B)". AON-B is SPC5001. Our roster handles this correctly — OLG002 is SPC5001 with alias Moisan2017_AON and a populated case-encoded sequence (TGctacaaaacCCA), and MSR089/MSR090 are its Moisan rows. There is no duplicate molecule. One cosmetic improvement: the alias could read Moisan2017_AON-B for exactness.

P5 — Access classification — ACCEPT, executed, and it paid off immediately

I accept the correction without reservation; my SOURCES_TO_ACQUIRE.md inferred paywalls from publisher error responses, which do not establish entitlement walls. Two independent vindications of Beebop's rule appeared within this round:

- **German's routed paper** reads isOpenAccess=N, inEPMC=N in Europe PMC and is **free on J-STAGE** (§P1). Had I trusted the PMC flag, I would have filed it as an access request.
- **Conversely, "free" repeatedly turned out not to mean "data-bearing":** the vupanorsen article is free and contains no renal numbers; the VALOR tofersen paper is free and contains no renal content at all. Open access and evidentiary value are independent properties, and the register should carry both.

And the error runs in both directions — a caution for whoever works the access list. I mis-stated paywalls; discovery tools mis-state openness. Semantic Scholar advertises an openAccessPdf for ENVISION (10.1056/NEJMoa1913147) at nejm.org/doi/pdf/10.1056/NEJMoa1913147?articleTools=true. That URL returns a **Cloudflare 403 interstitial** — it is the publisher's own paywalled PDF, not an open deposit. An aggregator's open-access flag is a claim to be tested, exactly like a 403 is. This surfaced because a cross-checking agent asserted ENVISION was freely available from an institutional repository; I tested the route and could not substantiate it, so ENVISION remains on the access list below.

Corrected taxonomy for every source previously called blocked is in §5.

2. Current inventory at commit b51003c

Separating **Beebop-recounted**, **my source-verified**, and **curator/reported** values as required.

Quantity	Value	Qualification
Measurement rows	246	recounted by both parties, agree
Roster entries	65	63 distinct molecules + 2 class aggregates (§P4)
human_clinical rows	42	recounted, agree
human_invitro rows	67	recounted, agree; 55/67 numeric readout, 61/67 dosed
animal_invivo / animal_invitro	56 / 81	recounted, agree (137 animal total)
Distinct identified trial keys	12	register status
Verified distinct human trials	2	register status at this commit; 3rd (PROMOVI) now evidenced but not yet promoted
Independent human cohorts	unknown	no experiment-level register exists; not established
Human laboratory experiments	unknown	row count is not experiment count; not established
Sequence coverage	55/63 molecules (87.3%)	field coverage; per-position source verification not established
Tested-material purity	0/65	verified unreported across all sources reviewed, including the 4 read this round
Confirmed clinical negatives	1	recommended to become 0 pending German (§P3)

Remaining qualification blockers: (a) all grades provisional pending German; (b) no clinical row survives as a modeling negative if §P3 is accepted; (c) per-position chemistry is source-resolved for only part of the roster; (d) tested-batch identity is nowhere established — reference identity is not batch identity; (e) 7 of 10 identified trials unread (*6 after Revision B*).

3. Prioritized acquisition plan

Expected yields are stated as **unknown** where not established, per the request.

Rank	Target	Expected gain	Dependency
1	Sefaxersen GOLDEN trial , PMC13553563, free	Renal safety of an IgAN ASO in a non-renal (geographic atrophy) population — breaks the §P3 confound for a kidney-targeted molecule. Expected usable observations: unknown, likely ≥1 clinical row	none; retrievable now
2	Morimura 2026 figure-level extraction (file in hand)	Up to 3 human-lab constructs × 4 readouts × dose–time; expected qualified rows: unknown pending figure digitization	none; needs anchor-checked extraction

Rank	Target	Expected gain	Dependency
3	Vendor CoA for HY-132586A, HY-132610	First tested-material purity values in the dataset, for 2 constructs. Not the trial lot — must be recorded as vendor-lot characterization, not batch identity	Oscar: is vendor-lot purity in scope?
4	Sefaxersen phase 2 IgAN , Kidney Int, PMID 41443406, OA=N	Renal endpoints in IgAN; expected: unknown	access needed (§5)
5	IONIS-FB-LRx phase 2 IgAN , NCT04014335, completed	Completed trial, renal endpoints; publication not yet located	search + access
6	Circulation 2022 supplement only (vupanorsen)	Quantitative renal safety tables; would move MSR079 from asserted to measured	403 — human browser
7	4 NEJM articles + supplements	Closes 4 of 7 unread trials	institutional access
8	Ionis patent sequence listings	ISIS113715, ISIS104838 sequences	none; not attempted
—	VALOR tofersen	withdrawn — read in full, zero renal content	n/a

Staged files, with provenance preserved (research_staging/): 4 primary sources, original filenames, source URLs, retrieval date 2026-10-02, HTTP codes and SHA-256 recorded in logs/acquisition_manifest.json. No validated observation was extracted from any inaccessible text.

Shared-source ownership: none of the four files retrieved this round overlaps another endpoint's access work. I have proposed corrections to RESEARCH_ACCESS_REGISTER.md in §5 rather than editing another endpoint's entries.

4. Twenty-resource coverage log

Search date **2026-10-02**. Machine-readable copy: research_staging/logs/search20_log.json. Blocked services are **not** marked searched. Counts from loose relevance matching are flagged, because a large hit count against an unfiltered query is not coverage.

#	Resource	Query	HTTP	Result	Status
1	PubMed (E-utilities)	oligonucleotide nephrotoxicity	200	148	searched
2	Europe PMC (REST)	same; plus 5 targeted compound queries	200	2,319	searched — primary productive route ; serves fullTextXML for the OA subset
3	OpenAlex	same	429	—	not searched — rate-limited on 5 attempts across both search and single-DOI endpoints. Not a paywall
4	Semantic Scholar	same	200	183,651	searched; count is loose relevance matching , not 183k relevant papers

#	Resource	Query	HTTP	Result	Status
5	ResearchRabbit	—	200	—	not searched — account sign-in, no public API. Not a content paywall
6	Undermind	—	200	—	not searched — paid service, no public API
7	Elicit	—	200	—	not searched — account/subscrip- tion, no public API
8	Consensus	—	200	—	not searched — account required, no public API
9	GEO	antisense ol- igonucleotid e kidney; + 23 compound names	200	100 / per-compound	searched — spot-checked and negative , see below
10	SRA	same	200	40	searched; no renal oligo-toxicity deposit identified
11	PRIDE	oligonucleot ide kidney	200	0	searched
12	ProteomeXchang e (PROXI)	oligonucleot ide	200	no relevant	searched
13	BioStudies	oligonucleot ide kidney; + 23 compounds	200	100,732	searched; headline count is loose matching ; per-compound quoted queries used instead
14	ArrayExpress (in BioStudies)	same	200	6,284	searched; same caveat
15	ClinicalTrials.gov (API v2)	4 queries incl. siRNA nephropathy, antisense gl omerulonephr itis	200	13 / 21 / 3 / 2 → 36 distinct NCTs	searched — highest-yield resource this round
16	CTD	mipomersen batch query	302	—	not searched — endpoint redirects to HTML; oligos are poorly represented as CTD chemicals
17	ICE (NIEHS)	portal	200	—	reachable; no endpoint applicability — curated assays are small-molecule oriented

#	Resource	Query	HTTP	Result	Status
18	ToxCast / CompTox	portal	200	—	reachable; no endpoint applicability — library excludes therapeutic oligonucleotides
19	Zenodo	oligonucleotide nephrotoxicity; + 23 compounds	200	1,155	searched; no renal oligo-toxicity dataset identified
20	Dryad	oligonucleotide kidney	200	2	searched; not relevant

Totals: 12 searched, 4 not searched (login/subscription), 2 technically blocked, 2 reachable but not applicable.

Material negative result, verified not assumed. The repository route is empty for this endpoint. Per-compound searches across GEO/SRA/BioStudies/Zenodo returned hits for 22 of 23 compounds, but spot-checking the most promising — inotersen, 13 GEO records, the drug with our strongest clinical nephrotoxicity signal — showed **all 13 are neural-progenitor expression samples with zero renal relevance**. Keyword presence in repository metadata is not endpoint relevance. Consequence: for oligonucleotide nephrotoxicity, **primary literature and journal supplements are the only productive acquisition route**, which is why §3 and §5 are weighted entirely toward papers and supplements.

Additional sources inspected beyond the twenty: J-STAGE (productive — §P1), DOI content negotiation, journal supplement endpoints at ahajournals.org, White Rose institutional eprints, and the ClinicalTrials.gov registry records listed below.

New leads from resource 15, absent from our dataset. These are **registry records, not verified trials**, and are not counted anywhere:

NCT	Drug	Phase / status	Why it matters
NCT05797610	Sefaxersen (RO7434656)	3, active	ASO for IgA nephropathy; molecule absent from dataset
NCT04014335	IONIS-FB-LRx	2, completed	Complement factor B ASO in IgAN; completed, so data should exist
NCT07271186	ALN-ANG3	2, recruiting	Diabetic kidney disease
NCT06989359	ADX-038	2, recruiting	Complement-mediated kidney disease
NCT00554359	I5NP	1, completed	Same molecule as teprasiran — extension-counting item, not a new drug
NCT03159416	inclisiran	1, completed	Dedicated renal-impairment PK study

5. Paper / supplement / data access table

Classified per the request's taxonomy. Prices and entitlements are **unverified** throughout — none was checked, and no purchase, subscription or credential was used.

Source	Identifier / link	Discovered via	Needed file	Gap addressed	Attempt (2026-10-02)	Outcome	Class	Priority
Morimura K, Takahashi E, Maeda H, Nishioka Y, Araki A, Mizumoto H, Jimbo Y (2026). <i>Predicting nucleic acid drug-induced nephrotoxicity using a 3D human renal proximal tubule spheroid model</i> . J Toxicol Sci 51(1):75–87	10.2131/jts.51.75; PMID 41500580	German evidence-watch memo via Beebop	full article	human lab chemistry+ characterization	J-STAGE _pdf/-char/en	200, 13 pp retrieved	free publisher route (PMC flag said OA=N)	done
Thielmann M, Corteville D, Szabo G, Swaminathan M, Lamy A, Lehner LJ, et al. (2021). <i>Teprasiran, a Small Interfering RNA, for the Prevention of Acute Kidney Injury in High-Risk Patients Undergoing Cardiac Surgery</i> . Circulation	PMC8487715; 10.1161/CIRCULATION.120.053029; NCT02610283	my Oct-1 reply	full text	trial verification	EPMC REST fullTextXML	200, 7,330 w	open access	done
Groothoff J, Sellier-Leclerc AL, Deesker L, Bacchetta J, Schalk G, Tönshoff B, et al. (2024). <i>Nedosiran Safety and Efficacy in PH1: Interim Analysis of PHYOX3</i> . Kidney Int Rep	PMC11068990; NCT04042402	my Oct-1 reply	full text	trial verification	EPMC REST fullTextXML	200, 6,919 w	open access	done
McDonald CM, Shieh PB, Abdel-Hamid HZ, Connolly AM, Ciafaloni E, Wagner KR, et al. (2021). <i>Open-Label Evaluation of Eteplirsin in Patients with Duchenne Muscular Dystrophy Amenable to Exon 51 Skipping: PROMOV1</i> . J Neuromuscul Dis	PMC8673535; NCT02555552	my Oct-1 reply	full text	trial verification	EPMC REST fullTextXML	200, 7,661 w	open access	done
Moisan A et al. (2017). <i>Inhibition of EGF Uptake by Nephrotoxic Antisense Drugs In Vitro...</i>	PMC5363415	existing design_source	full text	sequence recovery	EPMC REST fullTextXML	200, read — no sequences present	open access; data absent from source	done

Source	Identifier / link	Discovered via	Needed file	Gap addressed	Attempt (2026-10-02)	Outcome	Class	Priority
Jaffe GJ, Wyckoff CC, McCaleb ML, Barrett TD, Frazer-Abel A, Norris D, et al. (2026). <i>GOLDEN: Efficacy and Safety of Complement Factor B Antisense, Sefaxersen, in Geographic Atrophy</i> . Ophthalmol Sci	PMC13553563	resource 2 this round	full text + safety tables	non-renal-indication renal safety	not yet attempted	—	open access	1
Sato T, Fukase H, Ishida T, Karasawa A (2026). <i>A First-in-Japanese Phase 1, Double-Blind, Placebo-Controlled, Parallel-Cohort Study of Sefaxersen</i> . Clin Pharmacol Drug Dev	PMC13555670 (Europe PMC marks this record a duplicate pending deletion; the sibling record is PMID 42713733, OA=N)	resource 2	full text	renal safety, phase 1	not yet attempted	—	open access	2
Sefaxersen phase 2 IgAN	PMID 41443406, Kidney International	resource 2	article + supplement	IgAN renal endpoints	not yet attempted	—	OA=N, inEPMC=N — entitlement unverified	3
Vupanorsen TRANSLATE-TIMI 70 supplement	10.1161/CIRCULATIONAHA.122.059266	Oct-1 reply	supplement only	quantitative renal safety for MSR079	publisher suppl; PMC bin; EPMC supplementary Files (not OA)	403 / 404	openly licensed (CC BY-NC-ND 4.0 per Crossref), technically blocked — see Revision B	4
NEURO-TTR inotersen	PMC12611561; 10.1056/NEJMoA1716793	register	article + renal AE tables	our top signal	EPMC XML; PMC HTML; europepmc HTML	XML refused; reCAPTCHA ; Cloudflare	in EPMC, OA=N; technical block	5
van Poelgeest SPC5001	PMC4693495; 10.1111/bcp.12738	register	article	SPC5001 human data	same three routes	same	in EPMC, OA=N; technical block	6
ENVISION givosiran	10.1056/NEJMoA1913147; NC T03338816	register; route from cross-check	article	trial verification	EPMC (no deposit); Semantic Scholar PDF (Cloudflare 403); CORE 403; King's Research Portal 200	RETRIEVE D — see Revision B	free at institutional repository (version of record)	done

Source	Identifier / link	Discovered via	Needed file	Gap addressed	Attempt (2 026-10-02)	Outcome	Class	Priority
Donidalorsen phase 3	10.1056/NEJMoA2402478	register	article + supplement	trial verification	EPMC search	no PMC deposit	missing deposit; paywall unverified	8
OCEANa-DOSE olpasiran	10.1056/NEJMoA2211023	register	article + supplement	trial verification	EPMC search	no PMC deposit	missing deposit; paywall unverified	9
Mongersen phase 2	10.1056/NEJMoA1407250	register	article	trial verification	EPMC search	no PMC deposit	missing deposit; paywall unverified	10
Sandeliu 2020	PMID 33084520	existing design_source	sequence listing	OLG014/OLG015	not attempted	—	unknown	11
Ionis patent sequence listings	—	—	sequence listings	ISIS113715, ISIS104838	not attempted	—	unresolved citation	12
VALOR tofersen	White Rose eprints	Beebop register	—	—	retrieved and read previously	zero renal content	free; struck as valueless	—

Precise request to Oscar. Three items need only a person with a browser, not a purchase: the **vupanorsen Circulation supplement** (403), **NEURO-TTR inotersen** PMC12611561, and **van Poelgeest** PMC4693495 — all free-or-deposited and defeated by bot protection, not by entitlement. The four NEJM articles are the only items that may genuinely require institutional access, and whether they do is **unverified**.

Proposed corrections to RESEARCH_ACCESS_REGISTER.md (not applied — it is shared): PMC8487715 is listed as a free teprasiran route, but direct retrieval returns a reCAPTCHA interstitial; the working route is the Europe PMC REST fullTextXML endpoint, which should be recorded as the general method for the OA subset. Second, the register should carry an "evidentiary value" column beside access status — VALOR tofersen is free and worthless here, and that combination needs to be representable.

6. Access-barrier classification summary

- **Confirmed publisher paywall:** none. No entitlement wall was directly verified this round.
- **Service login / subscription (search tools, not content):** ResearchRabbit, Undermind, Elicit, Consensus.
- **Technical blocking:** OpenAlex (429, 5 attempts); CTD batch endpoint (302); PMC HTML and europepmc.org HTML (reCAPTCHA / Cloudflare) for PMC12611561 and PMC4693495; ahajournals.org supplement (403).
- **Missing repository deposit:** 4 NEJM articles — **does not prove a paywall**.
- **Missing supplement:** PMC supplementary directory 404 for PMC9047643.
- **Unresolved citation:** MSR064 "KARDIA_trials"; Ionis sequence listings.
- **Free and retrieved:** 5 sources (4 this round + VALOR previously).
- **Free but evidentially empty:** VALOR tofersen.

7. What needs whose approval

German — scientific adjudication:

- 1 MSR066, and the derived renal-indication rule in §P3; accepting it takes confirmed clinical negatives to **zero**.
- 2 Whether MSR046's value may be split between the PHYOX3 paper and the Rivfloza label, and which supplies the number.
- 3 Whether the PHYOX3 renal SAEs (AKI, kidney failure, pyelonephritis; investigator-assessed unrelated, PH1 disease manifestations) should be recorded at all, and if so how.

- 4 Whether OLG008 alone constitutes a defensible matched human↔animal comparison (carried over from Oct-1; unanswered).
- 5 Whether kidney_toxicity_monitored rows should remain measurements (carried over; unanswered).
- 6 Whether a vendor-lot certificate of analysis is admissible as characterization, given it is not the trial lot.

Oscar — implementation and scope:

- 1 Promotion of MSR040/PROMOVI to verified, taking verified trials 2 → 3.
- 2 The extension-counting convention, before any of tepasiran's other four NCTs is added.
- 3 A human-browser download for the three bot-walled free items in §5.
- 4 Whether the four NEJM papers justify institutional access.
- 5 Whether to add sefaxersen and IONIS-FB-LRx as new constructs, and whether the sequence-coverage denominator becomes 63 rather than 65.

No final classifier, label change, merge or release is unlocked by this round, and none was performed.

RESEARCH ROUND COMPLETE — AWAITING GERMAN'S ADJUDICATION AND OSCAR'S IMPLEMENTATION APPROVAL